

Foundation (Jalapeno)

- Q1.** What is the purpose of graphing multiple inequalities on the same coordinate plane?
- (A) To find the intersection point
(B) To identify overlapping solution regions
(C) To solve a system of linear equations
(D) To find the slope of a line
(E) To determine the equation of a line
- Q2.** Which of the following is an example of a linear inequality?
- (A) $x + 2y = 5$
(B) $x + 2y > 5$
(C) $x + 2y < 5$
(D) $x + 2y \geq 5$
(E) All of the above
- Q3.** What does the shaded region represent when graphing a system of linear inequalities?
- (A) The solution to a single inequality
(B) The solution to a system of inequalities
(C) The equation of a line
(D) The slope of a line
(E) The y-intercept of a line
- Q4.** How do you determine the direction of the inequality when graphing?
- (A) By the slope of the line
(B) By the y-intercept of the line
(C) By the sign of the coefficient of x
(D) By the sign of the inequality symbol
(E) By the axis of symmetry
- Q5.** What is the term for the region where the solutions to multiple inequalities overlap?
- (A) Feasible region
(B) Solution set
(C) Graph
(D) Equation
(E) Inequality
- Q6.** Which of the following systems of inequalities has a feasible region?
- (A) $x + y > 2$ and $x + y < 3$
(B) $x + y > 2$ and $x + y > 3$
(C) $x + y < 2$ and $x + y > 3$
(D) $x + y = 2$ and $x + y = 3$
(E) $x + y \geq 2$ and $x + y \leq 3$
- Q7.** How do you find the vertices of the feasible region?
- (A) By finding the intersection points of the lines
(B) By finding the slope of the lines
(C) By finding the y-intercept of the lines
(D) By graphing the individual inequalities
(E) By solving the system of equations
- Q8.** What is the purpose of identifying the vertices of the feasible region?
- (A) To find the solution to a single inequality
(B) To find the equation of a line
(C) To determine the direction of the inequality
(D) To find the maximum or minimum value of a function
(E) To identify the corner points of the feasible region

Open Ended Questions (FRQ)

- Q9.** Graph the system of inequalities $x + 2y > 4$ and $x - 2y < 2$ on the same coordinate plane. Identify the feasible region and its vertices.
- Q10.** Solve the system of inequalities $x + y \geq 3$ and $x - y \leq 1$. Find the vertices of the feasible region and determine the direction of the inequalities.

Chef's Tips

- Tip 1: Emphasize the importance of graphing multiple inequalities on the same coordinate plane.
- Tip 2: Remind students to identify the overlapping solution regions and feasible regions.
- Tip 3: Encourage students to interpret the results and determine the vertices of the feasible region.